
LECTURE NOTES

Lecture 4 - Regularization

AI506 — Advanced Machine Learning

Simon Holm
February, 2026



DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

Contents

Cross Entropy Loss	3
Softmax	3
KL Divergence	3
Overcoming overfitting with regularization	3
Different regularizers	3
Early Stopping	3
Use of a second training step	3
Making synthetic data	3
Injecting Noise	4
Creating new Models by Removing Units	4

Cross Entropy Loss

Measures how well one probability distribution approximates another.

$$H(p|q) = - \sum_i^K p(y_i) \cdot \log(q(y_i))$$

Softmax

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j^K e^{z_j}}$$

KL Divergence

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \cdot \log\left(\frac{P(x)}{Q(x)}\right)$$

Overcoming overfitting with regularization

1. Increase capacity until overfitting
2. Regularize model to reduce overfitting

Different regularizers

- $L^2 = \Omega(\theta) = \frac{1}{2} \|w\|_2^2$
 - $J_1(w; X, y) = \frac{\alpha}{2} w^T w + J(w; X, y)$
 - $\nabla_w J_1(w; X, y) = \alpha w + \nabla_w J(w; X, y)$
 - $w' \leftarrow (1 - \varepsilon \alpha) w - \varepsilon \nabla_w J(w; X, y)$
- $L^1 = \Omega(\theta) = \sum_i |w_i|$
 - $J_1(w; X, y) = \alpha \|w\|_1 + J(w; X, y)$
 - $\nabla_w J_1(w; X, y) = \alpha \text{sign}(w) + \nabla_w J(w; X, y)$

Early Stopping

Every time the error on the validation set improves, we store a copy of the model parameters. When the training algorithm terminates, we return these parameters rather than the latest set.

Use of a second training step

- Early stopping requires a validation set, so some training data is not fed to the model.
- To best exploit this extra data, one can perform extra training after the initial training with early stopping has completed.
- Two basic strategies for including the validation data:
 1. Reinitialize the model to the untrained state and re-train on training **and** validation sets, with the identified optimal hyperparameters — including how long to train (number of epochs).
 2. Continue training with the final model on the validation set (e.g. if dealing with very large models).

Making synthetic data

Generate new samples just by transforming inputs.

- Images are high-dimensional and include a variety of variations; many are easily simulated.
- Translating the images a few pixels can greatly improve performance.
- Rotating and scaling are also effective.

- It may be more difficult in other areas (e.g. language).
- Invariance: $S(T(I)) = S(I)$
- Equivariance: $S(T(I)) = T(S(I))$

Injecting Noise

To improve robustness, train with random noise applied to the inputs.

Creating new Models by Removing Units

Dropout trains an ensemble of subnetworks.

By using dropout we can average multiple subnetworks, ultimately using much the same theory as making multiple models and averaging them.